12.4 Independent in Pairs, Not All Together

WORKSHEET

The question

STEP III 2021, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

? **Ask yourself** What is $\mathrm{P}(X_{12} = 1)$? What is $\mathrm{P}(X_{12} = 1 \text{ and } X_{23} = 1)$?

STAGE 1 Pairwise independence

Covers: Part (i), independence

? **Ask yourself** What is $\mathrm{P}(X_{12} = 1)$? And $\mathrm{P}(X_{12} = 1 \text{ and } X_{23} = 1)$? Compare with the product.

STAGE 2 Mean and variance of the score

Covers: Part (i), the mean and variance

Pause and try The score is a sum of $\binom{n}{2}$ variables. Expectation always adds. When does the variance of a sum equal the sum of the variances? Is pairwise independence enough?

STAGE 3 The variance formula

Covers: Part (ii)

Ask yourself If the Y_i have mean zero, the variance of the sum is E of its square. Expand the square: how often does each product $Y_i Y_j$ appear?

STAGE 4 The odd-and-even game

Covers: Part (iii), not independent

Ask yourself Find the distribution of Z_{12} and its mean. When is $Z_{12}Z_{23}$ non-zero, and what is it then? Compare $\mathrm{E}(Z_{12}Z_{23})$ with $\mathrm{E}(Z_{12})\mathrm{E}(Z_{23})$.

STAGE 5 The variance

Covers: Part (iii), the variance

> **Pause and try** Use Part (ii). Which pairs of pairs contribute to the cross terms: those with no player in common, or those sharing exactly one? How many of each are there?

What's really going on

Independence of several events

Events $A_1, \ldots, A_n$ are *independent* if, for every subcollection, the probability of the intersection is the product of the probabilities. For three events this means four conditions: three for the pairs, and $\mathrm{P}(A \cap B \cap C) = \mathrm{P}(A)\mathrm{P}(B)\mathrm{P}(C)$. Sergei Bernstein gave the example in Figure 12.4.1 (in the full section online) in the 1920s to show that the pairwise conditions do not imply the triple one. (Conversely, the triple condition alone does not imply the pairwise ones either.) The matching events X_{12}, X_{23}, X_{13} in the question are exactly like Bernstein's coins.

Uncorrelated is not independent

Zero covariance, $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$, is weaker still than independence. STEP III 2000 Q14 (in the exercises) shows that for variables taking only two values the two notions coincide, but with three values they do not. A simple example: let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Then $\mathrm{E}(XY) = \mathrm{E}(X^3) = 0 = \mathrm{E}(X)\mathrm{E}(Y)$, yet Y is completely determined by X. Covariance measures only linear association.

Cheap randomness

Random numbers are expensive for a computer to generate. In 1979 Larry Carter and Mark Wegman showed that for many algorithms, such as *hashing*, which stores data at pseudo-random locations, pairwise independent random choices are just as good as fully independent ones. Pairwise independent values can be generated very cheaply: for example, $a + bk \pmod{p}$ for $k = 0, 1, \ldots, p-1$, with a and b random, gives p values any two of which are independent. The variance calculation of this

question is why it works: Chebyshev's inequality needs only variances, and variances need only pairs.

Try this next

1 **Two values.** Try STEP III 2000, Question 14 (step.maths.org/questions/00-s3-q14). It shows that for random variables taking only two values each, $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ forces independence, and asks for a counterexample with three values.

2 **A bigger team.** For $n = 7$ in part (i), find the mean and standard deviation of the score, and compare the example score 7 with them.

3 **Birthdays again.** Use indicators to find the mean and variance of the number of pairs sharing a birthday among k people (Section 9.9). Are the indicators pairwise independent?

4 **Pairwise but not triple.** Find three events, each of probability $\frac{1}{2}$, with $\mathrm{P}(A \cap B \cap C) = \frac{1}{8}$ but which are not pairwise independent.

Further reading

- **Geoffrey Grimmett & David Stirzaker, *Probability and Random Processes*** (Oxford University Press), the opening chapters on independence.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University Press). Pairwise independence and its uses in algorithms.
- **Joseph Blitzstein & Jessica Hwang, *Introduction to Probability*** (CRC Press; free online), on indicators and covariance.

The full guided solution is in Section 12.4:
https://beyondstep.keithmcnulty.org/ch12/pairwise-independence.html